

Cisco Provider Edge Router (PER) Alarm Resolution SOP (ALARM_4787)

Node: Cisco-PE-Delhi-NCR-33 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_4787

Description

This SOP outlines the steps to diagnose, resolve, and verify a major alarm (ALARM_4787) on the Cisco Provider Edge Router (PER) - Node ID: Cisco-PE-Delhi-NCR-33.

Potential Root Causes

Cause	Probability
High CPU utilization on the router leading to packet drops and subsequent ALARM_4787.	High
Misconfiguration of QoS policies affecting traffic forwarding.	Medium
Hardware issues with the router's interface or line card.	Low

Diagnostic Commands

```
show processes cpu
show ip traffic
show mpls traffic-eng
show ip cef
show cef detail
```

Resolution Steps

- Step 1: {'Step': 'Run the show processes cpu command to identify the process consuming the most CPU resources.', 'Action': 'Check the output for any unexpected CPU spikes and identify the process causing them.'}
- Step 2: {'Step': 'Verify that QoS policies are correctly configured and not affecting traffic forwarding.', 'Action': 'Run the show running-config command and verify that QoS policies are correctly set up.'}
- Step 3: {'Step': 'Check for any hardware issues with the router's interface or line card.', 'Action': 'Run the show interface command and verify that the interface is up and running correctly.'}
- Step 4: {'Step': 'Check the CEF (Cisco Express Forwarding) configuration and ensure that it is correctly set up.', 'Action': 'Run the show cef detail command to verify CEF configuration.'}
- Step 5: {'Step': 'Clear the ALARM_4787 and verify that it does not reoccur.', 'Action': 'Use the clear alarm command to clear the ALARM_4787 and verify that it does not reoccur.'}

Post-Check Verification

```
show alarm
show processes cpu
show ip traffic
show mpls traffic-eng
show cef detail
```